# COL761 Assignment 2 – Q2: Route Blocking for Reducing Forest Fire Spread

## Task 1: Optimization Formulation (5 marks)

**Problem setting.** The forest is modeled as a directed graph $G = (V, E, p)$ where $V = \{1, 2, \ldots, n\}$ are regions, $E \subseteq V \times V$ are directed spread routes, and $p : E \to (0, 1]$ assigns an ignition probability to each route. Let $A_0 \subseteq V$ be the set of initially burning regions. The fire evolves in discrete time steps following the given independent cascade-style dynamics. Let $A_\infty$ be the set of burned regions at termination.

Blocking a set of routes $R \subseteq E$ removes these edges, yielding the modified graph $G_R = (V, E \setminus R, p)$. Define the expected burned count after blocking $R$ as

$$\sigma(R) = \mathbb{E}\left[|A_\infty|\right].$$

Given a budget $k$, the goal is to choose exactly $k$ edges to block to minimize $\sigma(R)$.

**Combinatorial optimization formulation.**

$$\min_{R \subseteq E} \ \sigma(R) \qquad \text{s.t.} \qquad |R| = k.$$

**Binary decision-variable formulation.** Index each edge by $e \in E$ and introduce a binary variable

$$x_e = \begin{cases} 1, & \text{if route } e \text{ is blocked,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $R(x) = \{e \in E : x_e = 1\}$. Then the problem is

$$\min_{x \in \{0,1\}^{|E|}} \ \sigma(R(x)) \qquad \text{s.t.} \qquad \sum_{e \in E} x_e = k.$$

Since the spread is stochastic, $\sigma(\cdot)$ typically has no closed form and is estimated via Monte Carlo simulation.

## Task 2: NP-Hardness (10 marks)

**Claim.** The route blocking problem is NP-hard (hence the optimization version is NP-hard).

**Proof idea.** We prove NP-hardness by showing that a restricted deterministic special case of the problem is NP-hard. Specifically, set $p(e) = 1$ for all $e \in E$. Then the spread becomes deterministic: a node burns iff it is reachable from $A_0$ in $G_R$, and thus

$$\sigma(R) = \mathbb{E}[|A_\infty|] = |A_\infty| = |\mathrm{Reach}_{G_R}(A_0)|.$$

**Reduction from Minimum $t$-Union (NP-hard).** *Minimum $t$-Union* is: given a universe $U = \{u_1, \ldots, u_n\}$, a family of sets $\mathcal{S} = \{S_1, \ldots, S_m\}$, and integers $t$ and $B$, decide whether there exists an index set $I \subseteq \{1, \ldots, m\}$ with $|I| = t$ such that

$$\left|\bigcup_{i \in I} S_i\right| \leq B.$$

This problem is NP-hard.

**Construction (polynomial time).** Given an instance $(U, \mathcal{S}, t, B)$, build a directed graph $G = (V, E, p)$ as follows:

- Nodes:
$$V = \{s\} \ \cup \ \{v_1, \ldots, v_m\} \ \cup \ \{w_1, \ldots, w_n\},$$
where $v_i$ corresponds to set $S_i$ and $w_j$ corresponds to element $u_j$.
- Edges (all with probability 1):
$$(s, v_i) \in E \quad \forall i \in \{1, \ldots, m\},$$
and
$$(v_i, w_j) \in E \quad \Longleftrightarrow \quad u_j \in S_i.$$
- Initial burning set: $A_0 = \{s\}$.
- Budget: set $k = m - t$ (i.e., we block exactly $m - t$ routes).

**Key observation.** Blocking an edge $(s, v_i)$ completely prevents any influence from set $S_i$ (it disconnects $v_i$ and all its outgoing edges from the source $s$). Under $p \equiv 1$, if $(s, v_i)$ is unblocked then $v_i$ surely burns at time 1; if it is blocked then $v_i$ never burns.

Hence, any feasible blocking set $R$ of size $k = m - t$ leaves exactly $t$ unblocked edges among $\{(s, v_i)\}_{i=1}^{m}$. Let $I$ be the indices of these unblocked set-nodes; then $|I| = t$.

An element-node $w_j$ burns iff there exists an unblocked set-node $v_i$ with $u_j \in S_i$. Therefore, the set of burned element-nodes is exactly

$$\{w_j : u_j \in \bigcup_{i \in I} S_i\},$$

and the total burned nodes are

$$|A_\infty| = 1 \ + \ |I| \ + \ \left|\bigcup_{i \in I} S_i\right| = 1 + t + \left|\bigcup_{i \in I} S_i\right|.$$

Thus,

$$|A_\infty| \leq 1 + t + B \quad \Longleftrightarrow \quad \left|\bigcup_{i \in I} S_i\right| \leq B.$$

**Correctness of reduction.** There exists a collection of $t$ sets with union size at most $B$ iff there exists a blocking set $R$ of size $k = m - t$ such that $\sigma(R) = |A_\infty| \leq 1 + t + B$.

Since the reduction is polynomial-time, the decision version of route blocking is NP-hard, and therefore the optimization problem is NP-hard.

**Remark (also holds under the $h$-hop model).** In this construction, all nodes that can burn are within distance at most 2 from $s$ ($s \to v_i \to w_j$). Hence for any $h \geq 2$, the $h$-hop restriction does not change the process, and NP-hardness still holds.

## Task 3: Structural Properties of $\sigma(R)$ (10 marks)

We analyze $\sigma(R)$ as a set function of blocked edges $R \subseteq E$, where $\sigma(R)$ is the expected number of burned nodes after removing $R$.

### (1) Monotonicity (3 marks)

**Claim.** $\sigma(R)$ is **monotone non-increasing** in $R$: if $R \subseteq R'$, then $\sigma(R') \leq \sigma(R)$.

**Proof.** Fix any outcome of the randomness (i.e., fix which attempted ignitions succeed). Consider the spread process on $G_R$ and on $G_{R'}$ with $R \subseteq R'$. Since $E \setminus R' \subseteq E \setminus R$, the process on $G_{R'}$ has a subset of the available routes compared to $G_R$. Therefore, for this fixed outcome, the set of nodes that become reachable/ignited in $G_{R'}$ is a subset of those ignited in $G_R$, implying $|A_\infty(R')| \leq |A_\infty(R)|$ pointwise. Taking expectations over the randomness yields $\sigma(R') \leq \sigma(R)$. $\square$

### (2) Submodularity (7 marks)

Recall that a set function $f$ is submodular if for all $A \subseteq B$ and $e \notin B$,

$$f(A \cup \{e\}) - f(A) \;\geq\; f(B \cup \{e\}) - f(B).$$

For a minimization problem in $R$, it is often convenient to examine *marginal decrease*.

**Claim.** $\sigma(R)$ is **not submodular** in general.

**Counterexample (deterministic case).** Let $p(e) = 1$ for all edges. Consider nodes $s, a, b, t$ with $A_0 = \{s\}$ and edges:

$$(s, a),\ (s, b),\ (a, t),\ (b, t).$$

Intuitively, $t$ burns if at least one of $(a, t)$ or $(b, t)$ remains available and the corresponding parent ($a$ or $b$) burns.

Define two sets of blocked edges:

$$A = \emptyset, \qquad B = \{(s, a)\},$$

and let $e = (a, t)$ (note $e \notin B$). Compute burned counts:

- With $A = \emptyset$: all edges present, all nodes burn, so $\sigma(A) = 4$. Blocking $e$ removes $(a, t)$ but $t$ still burns via $b$, so $\sigma(A \cup \{e\}) = 4$. Thus the marginal change is $\sigma(A \cup \{e\}) - \sigma(A) = 0$.
- With $B = \{(s, a)\}$: node $a$ never burns. Remaining edges still burn $b$ and then $t$, so $\sigma(B) = 3$. Now block $e = (a, t)$ in addition. Since $a$ is already disconnected, blocking $(a, t)$ has no further effect: $\sigma(B \cup \{e\}) = 3$. Thus the marginal change is $\sigma(B \cup \{e\}) - \sigma(B) = 0$.

This instance yields equal marginals, so it does not yet violate submodularity. To obtain a strict violation, modify the graph so that $t$ can burn only through an intermediate node $c$ Add an extra node $c$ and edges

$$(a, c),\ (b, c),\ (c, t),$$

and remove $(a, t)$ and $(b, t)$. Now $t$ burns iff $c$ burns, and $c$ burns iff at least one of $a$ or $b$ ignites it. Choose blocking sets:

$$A = \{(s, a)\}, \quad B = \{(s, a), (s, b)\}, \quad e = (b, c).$$

Then in $G_A$, $b$ still burns so blocking $e$ prevents $c$ from being ignited (only route to $c$), causing $t$ to not burn: large decrease. In $G_B$, $b$ does not burn anyway, so blocking $e$ has no effect: zero decrease. Hence the marginal decrease of adding $e$ is *larger* for $B$ than for $A$ (violating diminishing returns), so $\sigma(\cdot)$ is not submodular.

**Conclusion.** $\sigma(R)$ is monotone non-increasing, but not submodular in general.

## Task 4: Algorithm Design (10 marks)

**High-level idea.** Since the exact optimization is NP-hard, we use a simulation-driven greedy strategy that iteratively selects the edge whose blocking yields the maximum estimated reduction in expected spread.

Define the *blocking gain*:

$$g(R) = \sigma(\emptyset) - \sigma(R),$$

the expected reduction in burned nodes compared to blocking nothing. Note $g(R)$ is monotone non-decreasing in $R$.

### Proposed algorithm (simulation-based greedy)

**Algorithm 1** Greedy Route Blocking with Monte Carlo Estimation

**Input:** Graph $G = (V, E, p)$, initial burning set $A_0$, budget $k$, number of simulations $M$
$R \leftarrow \emptyset$
**for** $i = 1$ to $k$ **do**
  **for** each $e \in E \setminus R$ **do**
    Estimate $\widehat{\sigma}(R \cup \{e\})$ via $M$ Monte Carlo simulations
  **end for**
  Choose $e^\star \in \arg\min_{e \in E \setminus R} \widehat{\sigma}(R \cup \{e\})$
  $R \leftarrow R \cup \{e^\star\}$
**end for**
**Output:** $R$

### Complexity (2 marks)

Let $m = |E|$ and let $T_{\text{sim}}$ be the time to run one fire-spread simulation (typically $O(m)$ worst-case, often smaller in practice, and $O(|E_h|)$ under the $h$-hop model). Each greedy round evaluates up to $m$ candidate edges, each requiring $M$ simulations. Thus the overall time is approximately

$$O\big(k \cdot m \cdot M \cdot T_{\text{sim}}\big).$$

### Approximation guarantee (3 marks)

If $g(R) = \sigma(\emptyset) - \sigma(R)$ were exactly **monotone submodular**, then the standard greedy algorithm would guarantee a $(1 - 1/e)$ approximation to the maximum gain (equivalently, it would produce a solution whose expected spread is within that factor of the optimal improvement).

However, Task 3 shows that $\sigma(R)$ is not submodular in general, so **a global** $(1-1/e)$ **guarantee cannot be claimed** for arbitrary instances of this problem. The proposed greedy algorithm should therefore be viewed as a practical heuristic that typically performs well, but without a universal approximation bound in the general case.

**Practical note.** Despite the lack of a universal guarantee, the algorithm is attractive because it directly optimizes the target objective (using simulation estimates), and it naturally applies to both the original spread model and the $h$-hop restricted model.

## Task 5: Competitive Route Blocking

### Objective

The objective of this task is to select a set of routes (edges) to block in a graph so that the expected number of burned regions after the spread of fire is minimized.

Let:

- $\sigma(\emptyset)$ be the expected number of burned regions when no routes are blocked.
- $\sigma(R)$ be the expected number of burned regions after blocking a set of routes $R$.
- $\rho(R)$ be the reduction ratio defined as

$$\rho(R) = \frac{\sigma(\emptyset) - \sigma(R)}{\sigma(\emptyset)}$$

The goal is to maximize $\rho(R)$ subject to the constraint that the number of blocked routes $|R|$ does not exceed the budget $k$.

### Approach

The algorithm identifies routes that contribute the most to the spread of fire from the initial seed regions. The exact greedy algorithm is computationally expensive for large graphs, therefore we use efficient heuristics to approximate the edge importance.

Two propagation settings must be handled:

- **Unlimited Spread (**$hops = -1$**):** Fire can propagate through the entire network.
- **Hop-Limited Spread (**$hops = h$**):** Fire spread is restricted to at most $h$ hops from the initial seed set.

The implementation adapts the strategy depending on the spread model.

- For the **unlimited spread case (**$hops = -1$**)**, a **layered BFS heuristic** is used. Nodes are explored in breadth-first order starting from the seed set, and edges closer to the seed nodes are prioritized since they have a higher probability of contributing to early propagation of the fire.

- For the **hop-limited case** ($hops > 0$), a **Monte Carlo simulation based approach** is used. Multiple stochastic simulations of the fire spread are performed to estimate the importance of each edge.

This hybrid strategy combines structural information from the graph with simulation-based influence estimation.

## Algorithm

The algorithm proceeds as follows.

**Step 1: Graph Loading**

The graph $G = (V, E, p)$ is loaded from the input file where each edge $(u, v)$ has an ignition probability $p$.

**Step 2: Seed Initialization**

The initial burning regions are read from the seed set file and used as the starting points for fire propagation.

**Step 3: Edge Importance Estimation**

- If $hops = -1$, a layered BFS traversal is performed starting from the seed nodes. Edges encountered earlier in the BFS layers are given higher priority since they lie on short propagation paths from the seed set.
- If $hops > 0$, Monte Carlo simulations of the fire spread process are executed. During each simulation, we track how frequently each edge contributes to successful fire propagation during the simulations. The influence score of an edge increases proportionally to its activation probability during the simulation.

**Step 4: Edge Ranking**

Edges are ranked according to their computed importance scores.

**Step 5: Edge Selection**

The top $k$ edges with the highest influence scores are selected as the blocked routes.

**Step 6: Periodic Output Writing**

To satisfy the assignment requirement for partial credit, the algorithm periodically writes the currently selected edges to the output file while running.

## Implementation Details

The implementation is provided in `fireblock.py` and executed using the wrapper script

```
bash forest_fire.sh <graph> <seedset> <output_file> <k> <r> <hops>
```

Important implementation features include:

- Efficient BFS-based traversal for layered edge selection.
- Monte Carlo simulation for estimating edge influence in hop-limited spread.
- Support for both unlimited and hop-limited propagation models.
- Periodic checkpoint writing to ensure partial credit if execution stops early.
- Fast runtime (typically less than 0.1 seconds for the provided datasets).

### Results on Public Datasets

Evaluation was performed using the provided script:

```
bash evaluate.sh <graph> <seedset> <blocked_routes> <k> <r> <hops>
```

#### Dataset 1

Parameters:

- $k = 50$
- simulations = 50
- $hops = -1$ (unlimited spread)

Results:

- $\sigma(\emptyset) = 36.2800$
- $\sigma(R) = 8.5000$
- Reduction = 27.78
- $\rho(R) = 0.765711$

#### Dataset 2

Parameters:

- $k = 20$
- simulations = 50
- $hops = 3$

Results:

- $\sigma(\emptyset) = 124.4200$
- $\sigma(R) = 15.2400$
- Reduction = 109.18
- $\rho(R) = 0.877512$

### Discussion

The algorithm achieves substantial reduction in expected burned regions.

Key observations include:

- Blocking edges close to the seed nodes significantly reduces the spread in the unlimited propagation case.
- Hop-limited spread allows Monte Carlo simulations to focus on a smaller neighborhood around the seed nodes.
- Combining structural heuristics with simulation-based scoring provides an efficient and effective strategy.

## Conclusion

We implemented a hybrid route-blocking strategy that combines BFS-based structural heuristics and Monte Carlo simulation to identify influential propagation routes. The approach efficiently selects high-impact edges while respecting the blocking budget.

The method achieves strong performance on the provided datasets with reduction ratios of 0.7657 and 0.8775 respectively.